

Linux Fundamentals - Class Cheat Sheet

Hexplot Alliance | Linux only | Most-used commands and concepts

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Mental model: know where you are -> move with cd -> inspect with ls -> create/copy/move files -> search with find/grep.

1. SSH + Navigation

Goal	Command
Connect to Concordia	ssh ENCS_USERNAME@login.encs.concordia.ca
Who am I?	whoami
Where am I?	pwd
List files/directories	ls
Detailed listing	ls -l
Include hidden files	ls -a
Detailed + hidden	ls -al
Enter directory	cd folder
Go up one directory	cd ..
Go home	cd ~
Previous directory	cd -

Paths/commands are case-sensitive. The \$ prompt is not part of the command.

2. Paths + Filesystem

Goal	Command
Filesystem root	/
Current directory	.
Parent directory	..
Home directory	~
Absolute path	/home/user/code/main.c
Relative path	code/main.c
Show directory tree	tree
Directories only	tree -d
Include hidden items	tree -a
Limit depth	tree -L 2

Absolute paths start at /. Relative paths start from your current directory.

3. Create / Copy / Move / Delete

Goal	Command
Create empty file	touch file.txt
Create directory	mkdir folder
Create parent dirs too	mkdir -p a/b/c
Copy file	cp source destination
Copy directory	cp -r source destination
Move file	mv source destination
Rename file	mv oldname newname
Delete file	rm file.txt
Delete empty directory	rmdir folder
Delete directory + contents	rm -r folder

Use cp -i or mv -i to ask before overwriting. Be careful with recursive deletion.

4. Help / Manual Pages

Goal	Command
Quick syntax help	command --help
Open manual	man command
Example manual	man ls
Search manuals by keyword	man -k keyword
man/less: next page	Space
man/less: previous page	b
Search forward	/pattern
Next / previous result	n / N
Quit	q

Linux Fundamentals - Working With Files

Reading, searching, shortcuts, and command-line efficiency

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Key distinction: find searches for files/paths. grep searches for matching text inside files.

5. Read + Search

Goal	Command
Print a small file	<code>cat file.txt</code>
Browse a long file	<code>less file.txt</code>
First 5 lines	<code>head -n 5 file.txt</code>
Last 5 lines	<code>tail -n 5 file.txt</code>
Find .txt files	<code>find . -type f -name "*.txt"</code>
Find ignoring case	<code>find . -type f -iname "*.TXT"</code>
Search text in one file	<code>grep -n "ERROR" logs/app.log</code>
Recursive text search	<code>grep -Rni -- "todo" docs src</code>

find = paths by name/type. grep = matching lines inside files.

6. Bash Shortcuts

Key	Action
Tab	autocomplete a name
Tab twice	show possible completions
Up / Down	browse command history
Ctrl+A / Ctrl+E	start / end of command line
Ctrl+U / Ctrl+K	cut backward / forward
Ctrl+W / Ctrl+Y	cut previous word / paste
Ctrl+C	interrupt current command
Ctrl+L	clear visible screen

7. History

Goal	Command
Show recent history	<code>history 10</code>
Repeat previous command	<code>!!</code>
Preview previous command	<code>!!:p</code>
Run history entry #42	<code>!42</code>
Search command history	<code>Ctrl+R</code>

Be careful: repeating a command repeats its side effects too.

8. Aliases

Goal	Command
Create alias	<code>alias ll='ls -alh'</code>
Another alias	<code>alias croot='cd ~/hexploit-linux'</code>
See what a name resolves to	<code>type ll</code>
Remove alias	<code>unalias ll</code>
Bypass an alias	<code>\ls</code>

Aliases last for the current shell unless added to ~/.bashrc.